COUNTERPARTY CREDIT RISK & CVA ENGINE · FINAL REPORT

Phases A–F: from swap prices to collateralized CVA

The engine values a 10-swap portfolio on a real US Treasury curve and revalues it on 10,000 simulated Hull-White rate paths. It turns those values into netted counterparty exposure, prices unilateral CVA from default probabilities, and measures how netting, a collateral agreement, and credit and market assumptions change the result. Every number is validated against a known relationship.

Valuation date **02 Oct 2026** Tests **145 passed in 9.70s** Seed **42**

Market data → Swap valuation → Rate simulation → Future MtM → Exposure → Default probability → CVA → Netting / collateral → Risk report

10k × 121 Hull-White paths × dates for 10 swaps, revalued in 0.9 s	**$213.3k** portfolio CVA, ± $1,530 (1 std error)	**−60%** CVA cut by netting (gross $535.8k)	**−92%** further cut by a zero-threshold CSA ($16.7k left)	**145 tests** all passing, 10 files

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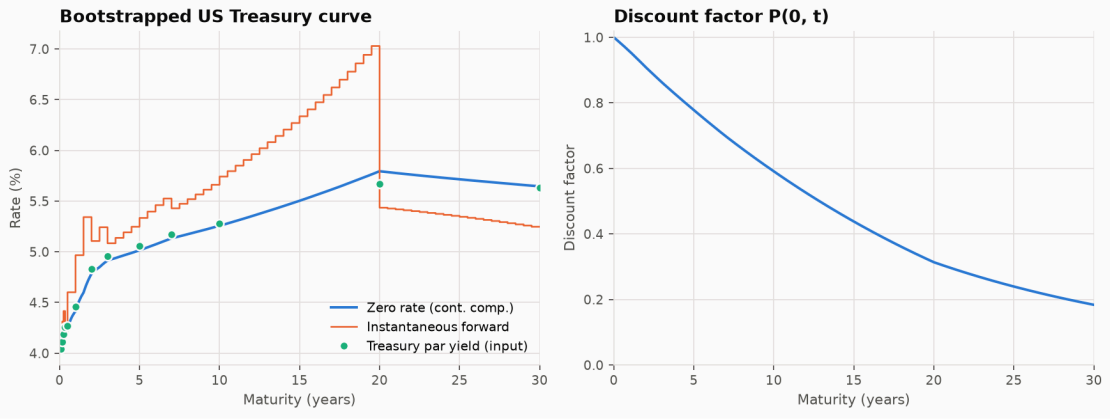
PHASE A · FOUNDATIONS

COMPLETE

What is a swap worth today?

- `scripts/fetch_treasury_curve.py` saved the Treasury par-yield curve for 02 Oct 2026; the snapshot is committed so every run is reproducible.
- `src/curves.py` bootstraps it into a zero curve and interpolates linearly in log discount factor.
- `src/instruments.py` and `src/swap_pricer.py` price vanilla fixed-for-floating swaps. The leg formulas take arrays of discount factors, so Phase B reuses them on simulated paths.
- `scripts/make_synthetic_portfolio.py` built a 10-trade book with fixed rates set off the par rate.

The curve



Zero rates rise from 4.03% at 1M to 5.79% at 20Y. The sawtooth forwards and the 20Y jump come from linear par-yield interpolation and the real 20Y/30Y inversion on this date. Within 10Y, where all trades mature, forwards stay between 4.03% and 5.74%.

Tenor	Par yield (input)	Zero rate, cont. comp.	P(0, T)
1M	4.04%	4.033%	0.99664
2M	4.11%	4.096%	0.99320
3M	4.19%	4.168%	0.98963
4M	4.26%	4.230%	0.98600
6M	4.27%	4.225%	0.97910
1Y	4.46%	4.413%	0.95683
2Y	4.83%	4.784%	0.90876
3Y	4.96%	4.914%	0.86293
5Y	5.06%	5.014%	0.77824
7Y	5.17%	5.133%	0.69816
10Y	5.28%	5.256%	0.59120
20Y	5.67%	5.795%	0.31382
30Y	5.63%	5.645%	0.18386

The portfolio

Values are from our side. PAY means we pay fixed and receive floating; fixed coupons are semiannual and floating coupons quarterly. NS_A mixes directions on purpose; NS_B is mostly payers.

Trade	Netting set	Fixed leg	Notional	Start → maturity	Fixed rate	Par rate	Fixed − par (bp)	MtM to us
A1	NS_A	PAY	$50m	0Y → 10Y	4.78%	5.28%	-50	$1,935,626
A2	NS_A	REC	$30m	0Y → 7Y	5.42%	5.17%	+25	$437,867
A3	NS_A	PAY	$25m	1Y → 6Y	5.27%	5.27%	+0	−$2,786
A4	NS_A	REC	$40m	0Y → 10Y	4.98%	5.28%	-30	−$929,100
A5	NS_A	PAY	$20m	0Y → 3Y	5.36%	4.96%	+40	−$221,073
A6	NS_A	REC	$15m	0Y → 2Y	5.43%	4.83%	+60	$170,012
B1	NS_B	PAY	$30m	0Y → 5Y	4.81%	5.06%	-25	$328,696
B2	NS_B	PAY	$20m	0Y → 8Y	5.41%	5.21%	+20	−$264,680
B3	NS_B	PAY	$25m	0Y → 4Y	4.61%	5.01%	-40	$359,402
B4	NS_B	REC	$10m	0Y → 3Y	5.26%	4.96%	+30	$82,902

Netting set	Counterparty	Trades	Gross notional	Net MtM	Sum of positive trade MtMs	Reduction from netting
NS_A	CPTY_A	6	$180m	$1,390,545	$2,543,504	45%
NS_B	CPTY_B	4	$85m	$506,321	$771,000	34%

Math to defend

```
# bootstrap: each quote is a bond priced at par
1 = y·Σ τₖ P(0,Tₖ) + P(0,Tₙ)    # bills: P = 1/(1+yT)
# interpolation: log P linear → flat forwards
A         = Σ τᵢ P(Tᵢ)
PV_fixed  = N·K·A
Fⱼ        = (P(Sⱼ)/P(Tⱼ) − 1) / τⱼ
PV_float  = N·Σ τⱼ Fⱼ P(Tⱼ) = N·(P(T₀) − P(Tₙ))
K_par     = PV_float / (N·A)
V         = sign·(PV_float − PV_fixed)
```

Checks

Worst Treasury bond repricing error	1.4e-14
A1 at its par rate	$0.000000
A1 payer / receiver	$1,935,626 / −$1,935,626
10Y par swap rate vs 10Y par yield	5.28% vs 5.28%

With one curve and semiannual fixed coupons, the par swap rate and the par bond yield are the same quantity, (1 − P(0,T)) / A.

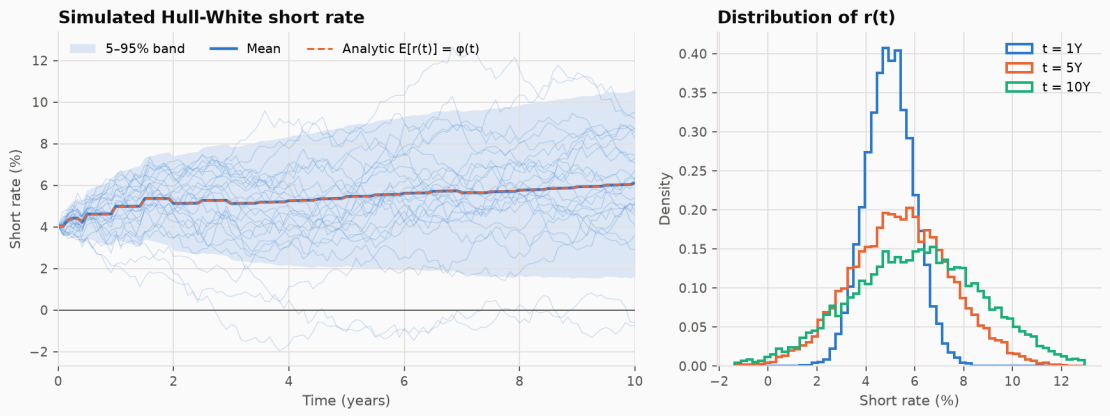
PHASE B · MARKET SIMULATION

COMPLETE

How could rates evolve, and what would the swaps be worth?

- `src/rate_models.py` : one-factor Hull-White closed forms for bond prices, the short rate and the stochastic discount factor.
- `src/simulation.py` : time grid (monthly plus every reset and payment date, 121 dates), exact path simulation and revaluation into a trades × paths × dates cube.

Hull-White fits today's curve exactly, which Vasicek cannot. Writing r(t) = x(t) + φ(t) leaves a zero-mean Ornstein-Uhlenbeck process with a known Gaussian transition, simulated exactly with no time-step error. Parameters are assumed, not calibrated: a = 3% per year and σ = 1%/√year, the Hull-White short-rate diffusion volatility. σ is a model parameter, not a quoted swaption normal (Bachelier) vol.



The simulated mean lies on the analytic E[r(t)] = φ(t). Hull-White is Gaussian, so rates can go negative (1.3% of paths at 10Y).

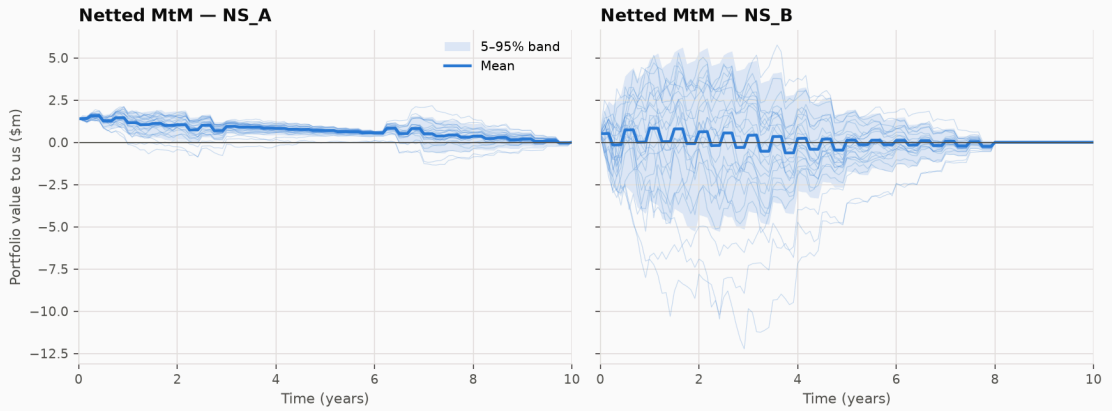
t	Mean (MC)	Mean (analytic)	Std (MC)	Std (analytic)	5–95% range	P(r < 0)
0.5Y	4.61%	4.60%	0.70%	0.70%	3.45% – 5.77%	0.00%
1Y	4.99%	4.97%	0.98%	0.99%	3.37% – 6.61%	0.00%
2Y	5.12%	5.13%	1.38%	1.37%	2.88% – 7.40%	0.02%
5Y	5.46%	5.44%	2.05%	2.08%	2.14% – 8.89%	0.36%
10Y	6.12%	6.12%	2.74%	2.74%	1.61% – 10.61%	1.28%

The martingale test

In an arbitrage-free model fitted to the curve, E[D(0,t)·P(t,T)] = P(0,T). In the 10,000-path base run below, the worst |z| is 1.62, inside the 4-standard-error test tolerance. A separate 400,000-path rerun on a coarse grid gives a worst |z| of 1.16, so the base-run deviations are sampling noise, not bias.

t	T	E[D(0,t)·P(t,T)] (MC)	P(0,T) (curve)	Std error	z
1Y	2Y	0.90854	0.90876	0.00014	-1.62
1Y	5Y	0.77771	0.77824	0.00033	-1.62
2Y	10Y	0.59122	0.59120	0.00067	+0.04
5Y	10Y	0.59052	0.59120	0.00089	-0.75
10Y	10Y	0.59096	0.59120	0.00097	-0.24

Revaluing the book on every path



Netted value to us by netting set. NS_A barely moves; NS_B swings with the level of rates.

Netting set	t	Mean netted MtM	5–95% range
NS_A	0.5Y	$1.26m	$0.87m to $1.63m
NS_A	1Y	$1.16m	$0.45m to $1.83m
NS_A	2Y	$1.04m	$0.19m to $1.85m
NS_A	5Y	$0.69m	$0.40m to $0.92m
NS_B	0.5Y	$0.73m	−$2.57m to $3.92m
NS_B	1Y	$0.84m	−$3.43m to $4.93m
NS_B	2Y	$0.62m	−$4.29m to $5.32m
NS_B	5Y	$0.12m	−$1.70m to $1.90m

Math to defend

```
dx = −a·x dt + σ dW,    r = x + φ(t)
φ(t) = f(0,t) + σ²/(2a²)·(1 − e^(−at))²
B(t,T) = (1 − e^(−a(T−t))) / a
P(t,T) = P(0,T)/P(0,t)·exp(−B·x − ½B²·Vₓ(t)
                           − B·σ²/(2a²)·(1 −
e^(−at))²)
Vₓ(t)  = σ²/(2a)·(1 − e^(−2at))
D(0,t) = P(0,t)·exp(−∫x ds − ½Var[∫x ds])
```

Revaluation rules

- P(t,T) is computed once per date for every future portfolio date; trades reuse the Phase A leg formulas.
- A floating coupon fixes on its reset date at L = (1/P(S,T) − 1)/τ on each path and keeps that fixing.
- V(t) excludes cash flows paid at t; a trade is worth zero from its final payment.
- With σ = 0, every path reproduces the forward value to about 1e-9 relative.

PHASE C · EXPOSURE

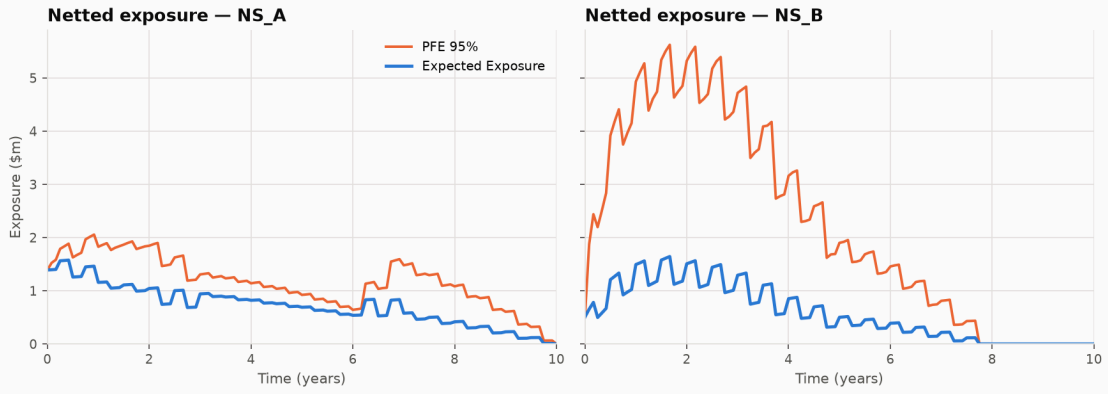
COMPLETE

How much would we lose if the counterparty defaulted?

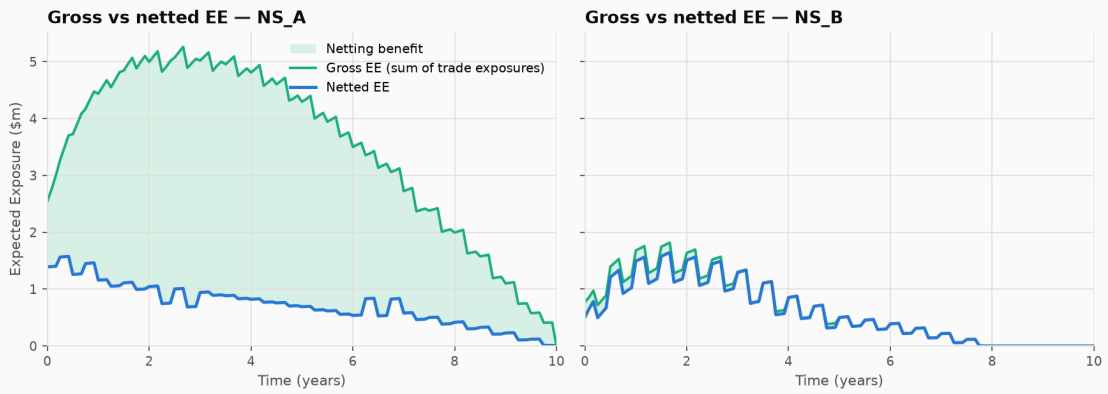
- `src/exposure.py` : netted and gross exposure per netting set, Expected Exposure, PFE and EPE, with summary and profile tables.
- `run_all.py` stage C writes the profiles and three figures; `notebooks/03_exposure_profiles.ipynb` walks through them.

At default we lose the netting set's value only when it is positive to us; a negative value is still owed. Trades in a netting set close out as one amount, so values are netted on each path and date before taking the positive part. PFE is the 95th percentile of exposure, taken as an actual simulated value (empirical inverse CDF). EPE here always means the time-average of EE over the netting set's whole life, not the regulatory Effective EPE.

Netting set	Life	EPE (time-avg EE) netted	EPE (time-avg EE) gross	Netting reduction	Peak EE	Peak PFE 95%
NS_A	10Y	$0.72m	$3.50m	79%	$1.58m at 0.4Y	$2.06m at 0.9Y
NS_B	8Y	$0.72m	$0.79m	8%	$1.64m at 1.7Y	$5.62m at 1.7Y

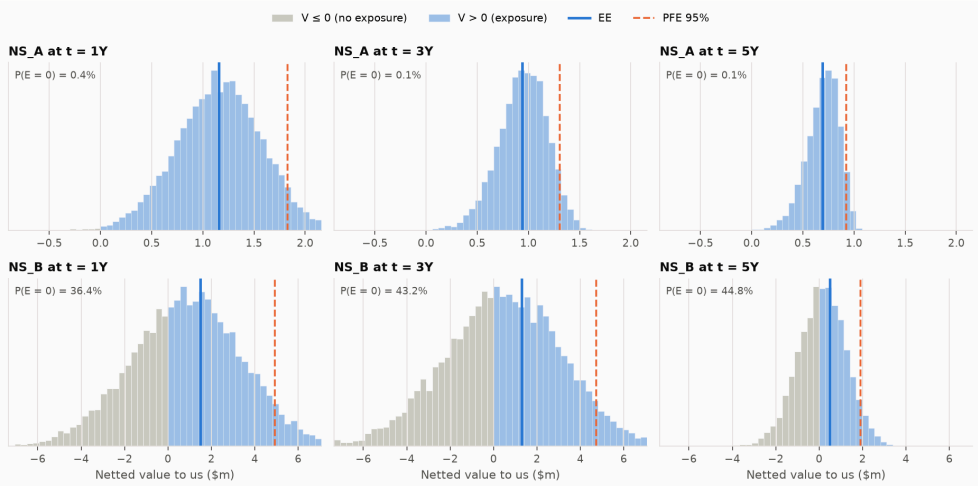


Netted EE and PFE at 95%. NS_B rises as rates spread out, peaks around 1.5–2Y and runs off. NS_A starts high because it is already in the money, then declines. The sawtooth comes from quarterly floating and semiannual fixed payments.



Gross EE (each trade on its own) against netted EE. Netting removes 79% of NS_A's time-average EE (EPE) but only 8% of NS_B's; once B4, NS_B's only receiver, matures at 3Y, gross equals netted.

Netting set	t	EE netted	PFE 95% netted	EE gross
NS_A	0.5Y	$1.26m	$1.63m	$3.72m
NS_A	1Y	$1.16m	$1.83m	$4.44m
NS_A	2Y	$1.04m	$1.85m	$5.00m
NS_A	3Y	$0.94m	$1.31m	$5.02m
NS_A	5Y	$0.69m	$0.92m	$4.29m
NS_A	7Y	$0.58m	$1.48m	$2.72m
NS_B	0.5Y	$1.21m	$3.92m	$1.40m
NS_B	1Y	$1.49m	$4.93m	$1.68m
NS_B	2Y	$1.51m	$5.32m	$1.64m
NS_B	3Y	$1.29m	$4.72m	$1.30m
NS_B	5Y	$0.50m	$1.90m	$0.50m
NS_B	7Y	$0.22m	$0.81m	$0.22m



Netted value at 1Y, 3Y and 5Y. Colored bars are paths with exposure; grey bars are paths where we would owe the counterparty. Vertical lines mark EE and PFE.

Math to defend

$$E_{net}(t) = \max(\Sigma_i V_i(t), 0)$$
$$E_{gross}(t) = \Sigma_i \max(V_i(t), 0) \geq E_{net}(t)$$
$$EE(t) = \text{mean over paths of } E(t)$$
$$PFE_q(t) = \min e : P_n(E(t) \leq e) \geq q$$
$$EPE = (1/T)\cdot\int_0^T EE(t)\,dt$$

EPE = time-average EE over the life; not Effective EPE

$\max(a+b,0) \leq \max(a,0)+\max(b,0)$: netting never adds exposure

Checks (base case, every path and date)

Smallest exposure	0.00
Largest netted − gross	0.00
$\lvert EE(0) - \max(V_0, 0)\rvert$	3.1e-07
PFE reconciles with empirical CDF	yes, every date
Paths with zero exposure, NS_B at 1Y / 3Y / 5Y	36% / 43% / 45%

PHASE D · CREDIT AND CVA

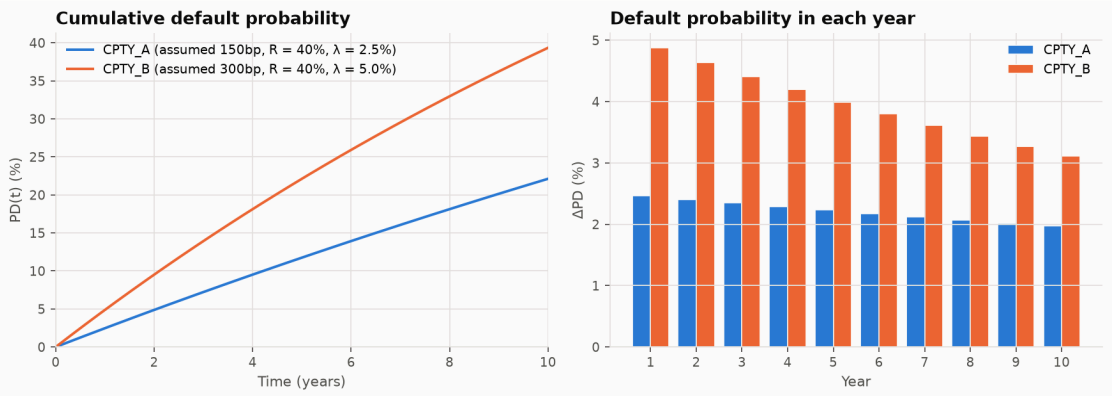
COMPLETE

What is the expected loss from that exposure?

- `src/credit.py`: hazard-rate survival curves (flat or piecewise) from assumed CDS-equivalent spreads and recoveries, using the credit-triangle approximation λ ≈ s/(1−R), not a CDS-curve calibration.
- `src/cva.py`: pathwise-discounted CVA as the average of a per-path CVA, so every estimate carries a standard error. It is integrated on the full simulation grid, with the deterministic curve-DF version as a benchmark.
- `src/validation.py`: credit sensitivities with the simulated exposures held fixed. `notebooks/04_cva_and_sensitivities.ipynb` walks through all of it.

The spreads in `counterparties.csv` are assumed counterparty spread scenarios (CDS-equivalent), not market quotes. λ ≈ s/(1−R) ignores premium accrual, discounting inside the CDS legs and the spread term structure. Hazard rates are deterministic and independent of rates, so there is no wrong-way risk.

Netting set	Counterparty	Assumed spread	Recovery	Hazard λ ≈ s/(1−R)	LGD	PD to last maturity
NS_A	CPTY_A	150bp	40%	2.5%	60%	22.1% (10Y)
NS_B	CPTY_B	300bp	40%	5.0%	60%	33.0% (8Y)

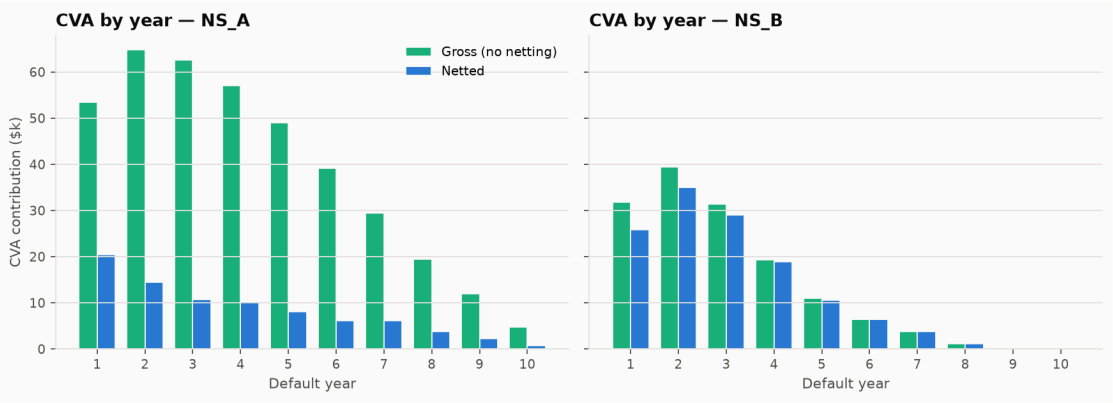


Each year's default probability shrinks slowly because the counterparty must first survive to that year.

CVA

Netting set	CVA netted (headline)	Std error	95% interval	CVA gross	Netting reduction	Curve-DF benchmark	Benchmark vs headline
NS_A	$82,715	± $208	$82,307 – $83,122	$391,613	79%	$83,974	+1.5%
NS_B	$130,591	± $1,348	$127,948 – $133,234	$144,217	9%	$134,015	+2.6%
TOTAL	**$213,306**	**± $1,530**	**$210,307 – $216,305**	**$535,830**	**60%**	**$217,989**	**+2.2%**

The headline discounts each path with its simulated bank account D(0,t), because exposure and discount factors are correlated under stochastic rates. The spec §7 approximation P(0,t)·EE(t) is kept as a benchmark. The integral runs over all 120 intervals of the simulation grid; the yearly buckets below only regroup it. The total's standard error comes from the per-path sum, since both netting sets share the same paths and their errors are correlated.



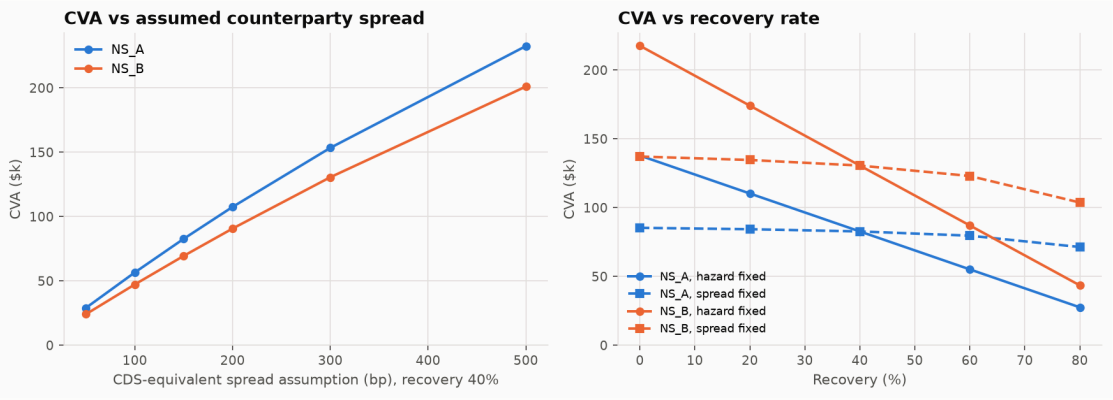
CVA by default year. NS_B's CVA sits in years 1–3, where its exposure peaks. NS_A's declines from the start; most of its gross CVA disappears with netting.

Netted CVA ($k)	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	Total
NS_A	20.5	14.5	10.8	10.2	8.1	6.1	6.1	3.7	2.2	0.7	**82.7**
NS_B	25.8	34.9	29.0	18.9	10.6	6.4	3.7	1.2	0.0	0.0	**130.6**

Robustness checks

Netting set	Headline (pathwise D)	Benchmark (curve DF)	Benchmark vs headline	Credit triangle s·∫dEE dt	Triangle vs headline
NS_A	$82,715	$83,974	+1.5%	$89,505	+8.2%
NS_B	$130,591	$134,015	+2.6%	$147,926	+13.3%

The curve-DF benchmark overstates CVA slightly. These books gain when rates rise, which is exactly when the bank-account discount factor is small, and the deterministic approximation ignores that covariance. The credit-triangle shortcut ignores that the counterparty must first survive, so it overstates CVA, more so for the riskier CPTY_B.



Credit sensitivities with exposures held fixed. CVA rises with the assumed spread, slightly less than proportionally. With the hazard fixed, CVA is exactly proportional to 1 − R; with the spread fixed, the hazard is re-implied and CVA barely moves.

View	Input	NS_A	NS_B
Assumed spread (bp), R = 40%	50bp	$29.0k	$24.1k
	100bp	$56.6k	$47.3k
	150bp	$82.7k	$69.4k
	200bp	$107.5k	$90.7k
	300bp	$153.4k	$130.6k
	500bp	$232.5k	$201.0k
Recovery, hazard fixed	0%	$137.9k	$217.7k
	20%	$110.3k	$174.1k
	40%	$82.7k	$130.6k
	60%	$55.1k	$87.1k
	80%	$27.6k	$43.5k
Recovery, spread fixed	0%	$85.3k	$137.2k
	20%	$84.3k	$134.6k
	40%	$82.7k	$130.6k
	60%	$79.6k	$123.0k
	80%	$71.4k	$103.7k

Math to defend

```
S(t)   = exp(−∫₀ᵗ λ(u) du)
ΔPDᵢ   = S(tᵢ₋₁) − S(tᵢ)
λ      ≈ s / (1 − R)      # credit-triangle approximation
CVA    = LGD·Σᵢ ½[dEE(tᵢ₋₁) + dEE(tᵢ)]·ΔPDᵢ
dEE(t)= E[D(0,t)·E(t)]   # benchmark: P(0,t)·EE(t)
       ⇒ CVA = mean over paths of CVA_path
SE     = std(CVA_path) / √N
```

Spec §11 checks

λ = 0 → CVA	exactly 0
R = 100% → CVA	exactly 0
Higher λ → higher CVA	tested, 7 levels
Higher R → lower CVA	exactly ∝ 1 − R
2 × notional → CVA	exactly 2×
SE ∝ 1/√N	tested, 1k/4k/16k paths

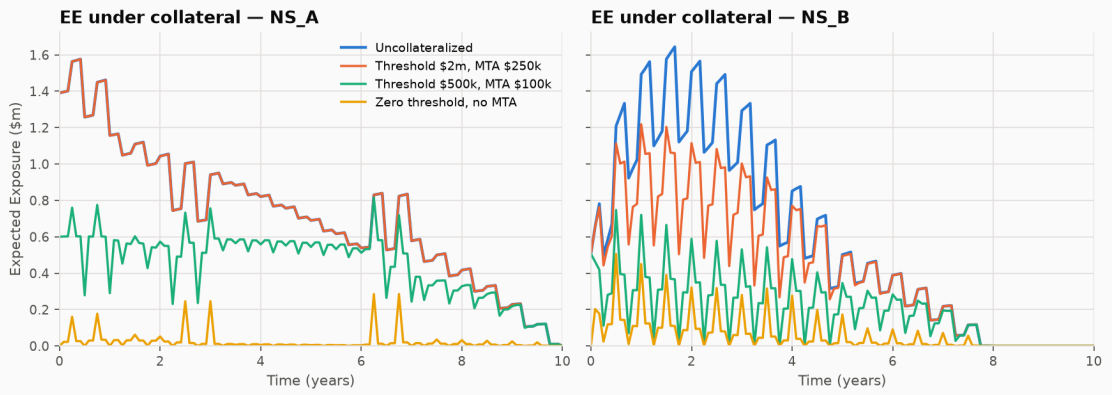
PHASE E · COLLATERAL

COMPLETE

How much do netting and collateral reduce CVA?

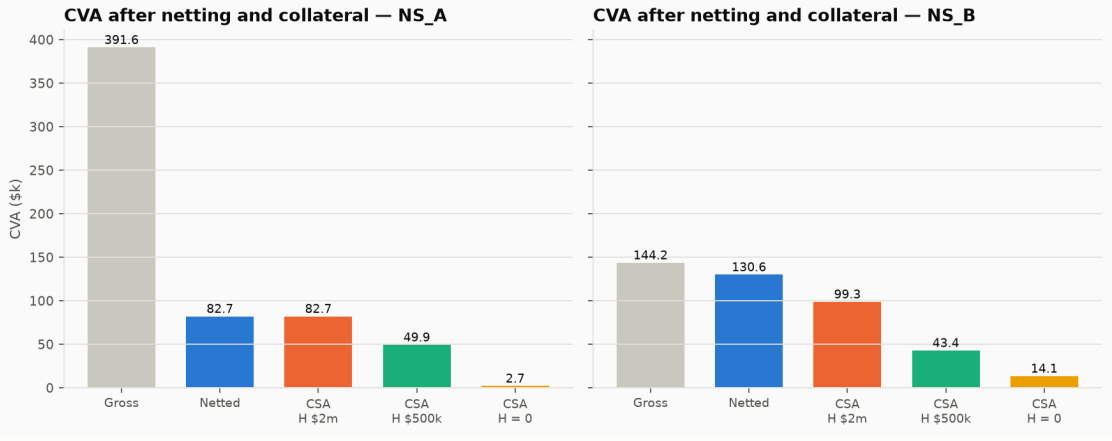
- `src/collateral.py` : a one-way CSA (the counterparty posts, we never do) with threshold H, minimum transfer amount and margin period of risk (MPoR), plus a shared analysis of every scenario.
- `src/simulation.py::refine_paths` : an exact Gaussian bridge adds the look-back dates t − MPoR to the Phase D paths without changing any existing date. `notebooks/05_collateral.ipynb` walks through it.

If the counterparty defaults at t, the last collateral call it met was at t − MPoR, so collateral held is C(t) = max(V(t − MPoR) − H − MTA, 0) and collateralized exposure is max(V(t) − C(t), 0). Collateral can only reduce exposure, a tighter threshold reduces it on every path, and an infinite threshold gives back the uncollateralized case. Simplifications: MTA is added to the threshold, both parties keep paying coupons during the MPoR, and there is no initial margin or haircut.



Netted EE under each CSA scenario. A $2m threshold barely touches NS_A, whose exposure rarely exceeds it. A $500k threshold caps NS_A near H + MTA from day one. Under a zero threshold, what remains is value moves inside the MPoR plus spikes on dates when we make a net coupon payment while the counterparty still holds collateral sized before it.

Netting set	CSA scenario (MPoR 2 weeks)	EPE (time-avg EE)	Peak EE	Peak PFE 95%	CVA ± std error	vs uncollateralized
NS_A	Uncollateralized	$0.72m	$1.58m	$2.06m	$82,715 ± $208	—
NS_A	Threshold $2m, MTA $250k	$0.72m	$1.58m	$2.06m	$82,698 ± $207	—
NS_A	Threshold $500k, MTA $100k	$0.45m	$0.82m	$1.10m	$49,909 ± $71	−40%
NS_A	Zero threshold, no MTA	$0.02m	$0.29m	$0.50m	$2,664 ± $3	−97%
NS_B	Uncollateralized	$0.72m	$1.64m	$5.62m	$130,591 ± $1,348	—
NS_B	Threshold $2m, MTA $250k	$0.56m	$1.22m	$3.30m	$99,306 ± $842	−24%
NS_B	Threshold $500k, MTA $100k	$0.25m	$0.75m	$2.04m	$43,449 ± $300	−67%
NS_B	Zero threshold, no MTA	$0.07m	$0.50m	$1.53m	$14,057 ± $87	−89%



From gross to netted to collateralized CVA. Netting does most of the work for the balanced NS_A; collateral does it for the one-directional NS_B.

Netting set	Stage	CVA	vs gross
NS_A	Gross (no netting, no collateral)	$391,613	—
NS_A	Netted, uncollateralized	$82,715	−79%
NS_A	Netted + CSA: Threshold $2m, MTA $250k	$82,698	−79%
NS_A	Netted + CSA: Threshold $500k, MTA $100k	$49,909	−87%
NS_A	Netted + CSA: Zero threshold, no MTA	$2,664	−99%
NS_B	Gross (no netting, no collateral)	$144,217	—
NS_B	Netted, uncollateralized	$130,591	−9%
NS_B	Netted + CSA: Threshold $2m, MTA $250k	$99,306	−31%
NS_B	Netted + CSA: Threshold $500k, MTA $100k	$43,449	−70%
NS_B	Netted + CSA: Zero threshold, no MTA	$14,057	−90%

Margin period of risk

Under a zero threshold, residual exposure depends on how long the value can move before collateral catches up. Pure diffusion would scale it like √MPoR (1.41× per doubling); the coupon spikes do not scale with the MPoR, so the observed growth is slower.

MPoR, zero threshold	NS_A CVA (share of uncollateralized)	NS_B CVA (share of uncollateralized)
7 days	$2,298 (2.8%)	$11,571 (8.9%)
14 days	$2,664 (3.2%)	$14,057 (10.8%)
28 days	$3,226 (3.9%)	$18,149 (13.9%)
Ratio per doubling	**1.16× / 1.21×**	**1.21× / 1.29×**

Math to defend

$$C(t) = \max(V(t - \mathrm{MPoR}) - H - \mathrm{MTA}, 0)$$

$$E_c(t) = \max(V(t) - C(t), 0) \le \max(V(t), 0)$$

look-back value on the same path, exact bridge:

$$Y \mid Z \sim N(\mu_Y + K(Z - A\mu_Y),\ S_1 - K A S_1)$$

$$K = S_1 A^{\top}(A S_1 A^{\top} + S_2)^{-1}$$

Checks

Uncollateralized CVA − Phase D headline	$0.00
CVA falls as collateral tightens	yes, both sets
Longer MPoR → more residual CVA	yes, both sets
Zero-threshold residual, NS_A / NS_B	3.2% / 10.8%

PHASE F · VALIDATION AND SENSITIVITIES

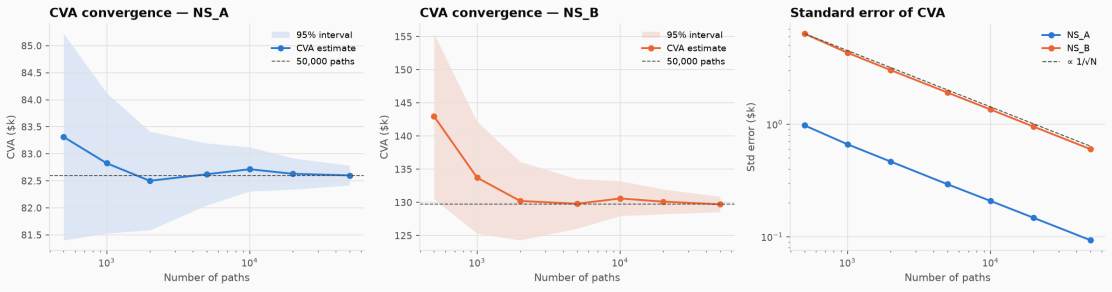
COMPLETE

How robust are the numbers?

- `src/validation.py`: Monte Carlo convergence to 50,000 paths; market sensitivities (σ, mean reversion, tenor, notional) under common random numbers; a tornado that adds spread and recovery.
- `src/reporting.py` writes `reports/final_project_summary.md` and the README results block from the generated tables. `docs/technical_note.md` documents every formula and model choice.

Monte Carlo convergence

Independent chunks of 10,000 paths are pooled; chunk 0 is the base case, so the 10,000-path point is the headline exactly.

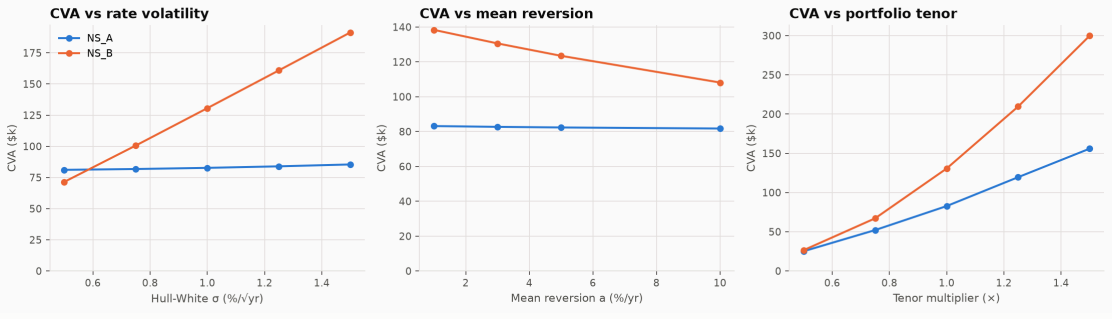


Estimates settle inside the earlier intervals and the standard error follows the $1/\sqrt{N}$ line. At 50,000 paths the portfolio CVA's relative standard error is 0.32%.

Paths	Portfolio CVA	Std error	95% interval	Relative std error
500	$226,283	$7,220	$212,132 – $240,434	3.19%
1,000	$216,557	$4,882	$206,988 – $226,125	2.25%
2,000	$212,717	$3,429	$205,997 – $219,437	1.61%
5,000	$212,414	$2,164	$208,173 – $216,655	1.02%
10,000	$213,306	$1,530	$210,307 – $216,305	0.72%
20,000	$212,744	$1,079	$210,629 – $214,858	0.51%
50,000	$212,305	$680	$210,972 – $213,639	0.32%

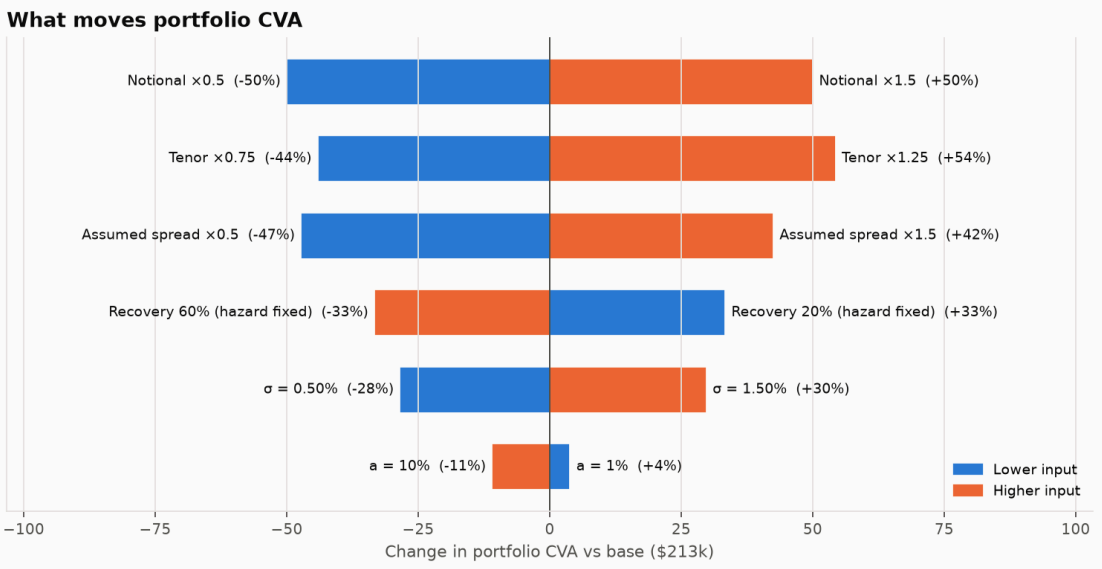
Market and portfolio sensitivities

Each run re-simulates with the base seed (common random numbers). For tenor, every trade is stretched and re-struck at par plus its original offset, so moneyness does not change. Notional scaling is exact because values are linear in notional.



NS_B's CVA rises almost linearly with σ because its exposure is option-like; NS_A barely moves because its exposure is mostly value it already holds. Higher mean reversion damps long-dated variance. Tenor is the strongest market effect, and convex.

Input	Value	NS_A	NS_B	Portfolio
Hull-White σ	0.50%	$81.1k	$71.4k	$152.6k
	0.75%	$81.8k	$100.6k	$182.4k
	1.00%	$82.7k	$130.6k	$213.3k
	1.25%	$83.9k	$160.9k	$244.8k
	1.50%	$85.5k	$191.4k	$276.8k
Mean reversion a	1%	$83.2k	$138.4k	$221.5k
	3%	$82.7k	$130.6k	$213.3k
	5%	$82.4k	$123.5k	$205.8k
	10%	$81.8k	$108.2k	$189.9k
Tenor multiplier	×0.5	$25.1k	$26.5k	$51.6k
	×0.75	$52.1k	$67.1k	$119.2k
	×1	$82.7k	$130.6k	$213.3k
	×1.25	$119.6k	$209.6k	$329.2k
	×1.5	$155.9k	$299.9k	$455.8k



Each input moved alone from the base portfolio CVA of $213.3k. Among the modelling inputs, portfolio tenor and assumed spread have the widest ranges.

Input	Lower value	Higher value	Range
Notional	×0.5: -50%	×1.5: +50%	100%
Portfolio tenor	×0.75: -44%	×1.25: +54%	98%
Assumed spread	×0.5: -47%	×1.5: +42%	90%
Recovery (hazard fixed)	20%: +33%	60%: -33%	67%
Rate volatility σ	0.50%: -28%	1.50%: +30%	58%
Mean reversion a	1%: +4%	10%: -11%	15%

Method

```
estimate(N) = mean of first N per-path CVAs
SE(N)       = std / √N
# chunk k uses seed 42 + 1000k; chunk 0 = base
# tenor ×k: T' = S + k(T − S), K' = K_par(T') + (K − K_par(T))
# σ change: same shocks, x scales linearly in σ
```

Checks

10,000-path point − headline	$0.00
50,000-path estimate inside 10,000-path 95% CI	yes
SE(10,000) / SE(50,000) vs √5	2.25 vs 2.24
CVA ↑ σ, ↑ tenor, ↓ mean reversion	yes

WHAT THE RESULTS SAY

Findings

Directional balance drives netting.

NS_A's payers and receivers offset, so netting cuts its time-average EE (EPE) by 79% and its CVA by 79%. NS_B is nearly all payers: 8% and 9%. Trade count has nothing to do with it.

Same EPE, very different risk.

Both sets have a time-average EE (EPE) near $0.72m. NS_A's exposure is steady and almost always positive; NS_B's is zero on 36%–45% of paths but has a far fatter tail (peak PFE $5.62m vs $2.06m).

Credit quality dominates the CVA ranking.

NS_B's CVA ($130.6k) exceeds NS_A's ($82.7k) because its assumed spread is twice as wide. At an equal 150bp spread, NS_A would carry more CVA ($82.7k vs $69.4k) because its exposure lasts longer.

Recovery matters less than it looks.

With the spread fixed, moving recovery from 0% to 80% changes NS_B's CVA only from $137.2k to $103.7k: a higher recovery implies a higher hazard, and the two largely cancel.

Netting and collateral solve different problems.

Netting removes 79% of NS_A's CVA but only 9% of NS_B's; a zero-threshold CSA then takes NS_B from $130.6k to $14.1k. Together they cut portfolio CVA from $535.8k to $16.7k.

Collateral never removes all the risk.

Even with no threshold, value moves inside the two-week MPoR and coupon payments inside the window leave 11% of NS_B's CVA; a four-week MPoR leaves more.

Tenor and spread move CVA most.

Over the tested ranges, portfolio tenor (98% range) and assumed spread matter most, then recovery (hazard fixed) and rate volatility σ; mean reversion matters least.

The base case is precise enough.

At 10,000 paths the portfolio CVA's standard error is 0.7%, far below every sensitivity, and the 50,000-path estimate lies inside its 95% interval.

A one-factor model flatters hedged books.

All rates move together, so NS_A's curve risk, and therefore its exposure and CVA, is understated.

Check noise against bias with more paths.

One mean-reversion regression landed 3.1 standard errors out at 10,000 paths; with 400,000 paths every regression was within 0.9.

VALIDATION COVERAGE

145 tests, 10 files

Exact formulas are tested to 1e-9 or tighter. Monte Carlo checks use fixed seeds and a tolerance of 4 standard errors that is never tuned to pass.

File	Tests	What it proves
`tests/test_curves.py`	11	Nodes reproduced, DF positive and decreasing, every Treasury bond reprices to par, closed-form flat-curve cases
`tests/test_instruments.py`	7	Schedules (regular, forward-starting, short stub), validation, portfolio loading and grouping
`tests/test_pricing.py`	19	Par swap = 0, payer = −receiver, float leg = $N(P(T_0) - P(T_n))$, closed-form par rate, generic aggregation
`tests/test_rate_models.py`	22	t = 0 reproduces curve, analytic moments of x and $\int x$, mean-reversion slope, martingale tests, common random numbers
`tests/test_simulation.py`	23	Grid contains every reset and payment date, cube(t = 0) = pricer, σ = 0 equals forward values, swap martingales, exact bridge
`tests/test_exposure.py`	15	Exposure ≥ 0, netted ≤ gross on every path, PFE reconciles with the empirical CDF, $EE(0) = \max(V_0, 0)$, N(0,1) case
`tests/test_credit.py`	12	Survival falls, PD rises, marginal PDs sum to PD(T), piecewise hazards, credit triangle, input loading
`tests/test_cva.py`	14	λ = 0 and R = 1 give zero CVA, monotone in λ / R / notional, netted ≤ gross, continuous closed form, $SE \propto 1/\sqrt{N}$
`tests/test_collateral.py`	11	Infinite threshold = uncollateralized, collateral never adds exposure, tighter threshold / smaller MTA / shorter MPoR lower exposure and CVA
`tests/test_validation.py`	11	Convergence chunk 0 = base run, $SE \propto 1/\sqrt{N}$, CVA rises with σ and tenor, falls with mean reversion, linear in notional

IN BRIEF

Project summary

> Built a Monte Carlo counterparty-credit-risk engine in Python for a 10-swap USD interest-rate portfolio: bootstrapped a US Treasury zero curve, simulated an exact one-factor Hull-White model (10,000 paths × 121 dates, martingale-tested) and priced unilateral CVA with pathwise discounting to ±0.7% Monte Carlo precision.

> Quantified counterparty-risk mitigation: legal netting cut portfolio CVA by 60% (79% for a balanced netting set vs 9% for a one-directional one), a zero-threshold CSA with a 14-day margin period removed a further 92%, and common-random-number sensitivities ranked the CVA drivers (portfolio tenor, credit spread, recovery, rate volatility).

MODEL RISK

Limitations

- **Single curve:** Treasury rates discount and project; there is no SOFR/OIS basis.
- **One-factor Gaussian rates:** all rates move together, so curve-hedged books such as NS_A look less risky than they are, and rates can go negative.
- **Uncalibrated rate parameters:** a and σ are assumptions, and CVA for directional books scales roughly linearly with σ.
- **Approximate credit:** hazards come from assumed spreads via the credit triangle, are deterministic and independent of rates (no wrong-way risk), and CVA is unilateral.
- **Simplified collateral:** MTA is added to the threshold, coupons are paid during the MPoR, and there is no initial margin or haircut.
- **Simplified schedules:** there are no day counts, calendars or historical fixings.

Every number in this report was read from the project's generated tables (`reports/tables`) or computed by its `src` code when the report was built; none were typed by hand. Figures are the outputs of `run_all.py`.